



Using machine learning to predict opioid misuse among U.S. adolescents

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ABSTRACT

This study evaluated prediction performance of three different machine learning (ML) techniques in predicting opioid misuse among U.S. adolescents. Data were drawn from the 2015–2017 National Survey on Drug Use and Health ($N = 41,579$ adolescents, ages 12–17 years) and analyzed in 2019. Prediction models were developed using three ML algorithms, including artificial neural networks, distributed random forest, and gradient boosting machine. The performance of the ML prediction models was compared with performance of the penalized logistic regression. The area under the receiver operating characteristic curve (AUROC) and the area under the precision-recall curve (AUPRC) were used as metrics of prediction performance. We used the AUPRC as the primary measure of prediction performance given that it is considered more informative for assessing binary classifiers on imbalanced outcome variable than AUROC. The overall rate of opioid misuse among U.S. adolescents was 3.7% ($n = 1521$). Prediction performance was similar across the four models (AUROC values range from 0.809 to 0.815). In terms of the AUPRC, the distributed random forest showed the best performance in prediction (0.172) followed by penalized logistic regression (0.162), gradient boosting machine (0.160), and artificial neural networks (0.157). Findings suggest that machine learning techniques can be a promising technique especially in the prediction of outcomes with rare cases (i.e., when the binary outcome variable is heavily lopsided) such as adolescent opioid misuse.

1. Introduction

Opioid misuse has been considered as one of the most serious threats to public health in the United States (Centers for Disease Control and Prevention, 2017). Among 47,600 overdose deaths involving opioids, over 35% deaths involved a prescription opioid (Scholl et al., 2019). The Substance Abuse and Mental Health Services Administration (SAMHSA) defines opioid misuse as misuse of prescription opioid pain relievers or use of heroin (Substance Abuse and Mental Health Services Administration, 2018). In 2017, there were 11.4 million (4.2%) past-year opioid misusers aged 12 or older in the United States, and the vast majority of the misused cases were prescription pain relievers (Substance Abuse and Mental Health Services Administration, 2018). There is also an increased awareness of the opioid misuse problem among adolescents, as a history of prescription opioid use in adolescence is a well-known critical predictor of other illicit drug use in young adulthood (Carlson et al., 2016; Cerdá et al., 2015; Compton et al., 2016). What is also concerning is that adolescent opioid misuse is positively correlated with mental health problems, poor academic performance, and risky behavior (Havens et al., 2011; Young et al., 2012). According to the most recent National Survey on Drug Use and Health

(NSDUH) data, over 3% of U.S. adolescents aged 12 to 17 years misused opioids in the past year (Center for Behavioral Health Statistics and Quality, 2018).

In light of opioid epidemic, a number of prior studies have sought to predict opioid misuse based on conventional statistical methods (Compton et al., 2018; Miech et al., 2015; Monnat and Rigg, 2016). Multivariable logistic regression model is the most commonly used method to identify significant predictors of opioid misuse. However, performance of logistic models in predicting opioid misuse has been rarely evaluated and compared with performance of other alternative prediction models. Machine learning (ML) is an emerging field of data analytic methods for predictive analysis and has rarely been used for opioid misuse prediction. ML approaches can be more accurate than conventional logistic regression models in many fields (Fernández-Delgado et al., 2014; Pinsky and Dubrawski, 2014). The ML offers a potential approach to build reliable and flexible predictive models by analyzing big data for algorithm development (Obermeyer and Emanuel, 2016) which is based on computational learning and pattern recognition. Such computerized algorithm development takes into account all possible interactions including nonlinear interactions and helps minimize errors between observed and predicted outcomes

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(Dreiseitl and Ohno-Machado, 2002).

The problem of highly skewed (i.e., imbalanced) data is widely observed in many domains (Japkowicz and Stephen, 2002), particularly in substance use research (Hinnant et al., 2015; Loughran et al., 2018; Nadel and Thornberry, 2017). Although opioid misuse is a widespread crisis, it still accounts for a small fraction of adolescent population. The conventional model such as the logistic regression model suffers from a low prediction capacity because of a high likelihood of the Type II error when the binary outcome variable is heavily lopsided (Wiles et al., 2006). However, in many cases, this problem is not fully identified and addressed despite its negative impact on model performance. While predictive power can be decreased with ML methods too if data are highly imbalanced (Krawczyk, 2016), the literature indicates that ML techniques may show good predictive performance even in highly imbalanced data (Alghamdi et al., 2017; Sakr et al., 2018; Chiew et al., 2019). Some ML methods are also known to mitigate the problems arising from multicollinearity (Altman and Krzywinski, 2018; Goin et al., 2018) while multiple logistic regression models are sensitive to such problems.

Despite the potential advantages of using ML techniques, few studies have employed these techniques to develop prediction models in the field of opioid dependence. Ahn and colleagues (Ahn et al., 2016; Ahn and Vassileva, 2016) used ML methods to identify profiles of individuals who were addicted to cocaine, heroin or amphetamine among Bulgarian adults and showed a high degree of predictive accuracy. A few other studies analyzed social media data using ML techniques to develop strategies to identify and analyze behaviors associated with opioid use in the general population although such studies are often limited by incomplete and shallow data (Kalyanam et al., 2017; Mackey et al., 2018). To date, no studies have used ML techniques to develop a prediction model of opioid misuse among adolescents, let alone in a nationally representative sample of U.S. adolescents.

The aim of this study was (1) to evaluate opioid misuse prediction performance of three different ML techniques, including artificial neural networks, distributed random forest, and gradient boosting machine, and (2) to compare such performance with the performance of the logistic regression in a nationally representative sample of U.S. adolescents ages 12–17 years. In terms of differential predictive power among the ML methods, we did not have a priori hypothesis because there is no clear consensus regarding the best choice of ML algorithm in terms of prediction performance (Thottakkara et al., 2016). This study would serve as an exploratory investigation as to what ML method may produce better performance in predicting opioid misuse in adolescents.

2. Material and methods

2.1. Data and study sample

The data for this study were drawn from three years (2015–2017) of the National Survey on Drug Use and Health (NSDUH). Of the 170,319 respondents who participated in the NSDUH 2015–2017, analyses in this study were restricted to 41,579 adolescents aged 12–17 years. The NSDUH is an annual survey of the civilian, noninstitutionalized population of the United States aged 12 years or older (Substance Abuse and Mental Health Services Administration, 2018). The survey is sponsored by SAMHSA within the U.S. Department of Health and Human Services and is a primary source for information on illicit drug use, alcohol use, substance use disorders, and mental health issues of the United States population. The NSDUH collects data using audio computer-assisted self-interviewing in which respondents read or listen to the questions on headphones, then enter their answers directly into a NSDUH laptop computer (Substance Abuse and Mental Health Services Administration, 2018). This method is believed to increase accurate reporting of information by providing respondents with a highly private and confidential mode for responding to questions. The NSDUH employs a stratified, multistage area probability sample that is designed to

be representative of the nation, resulting in differential probabilities of sample selection. To account for the complex sampling design of the NSDUH, sampling weights for the NSDUH were applied to all analyses. Given that the three years of combined data were used, we created a new weight by dividing the original weight by three (i.e., the number of data sets combined) in the pooled dataset (Weinberger et al., 2019).

For this study, only the data collected since 2015 were included in analysis because the definition of misuse of prescription drugs in NSDUH was changed in 2015. That is, we wanted to avoid the instrumentation bias that may arise when combining data with different operational definitions of opioid misuse. Before 2015, respondents were asked about the use of prescription drugs that were not prescribed for them or that respondents took only for the experience or feeling the drugs caused. The revised definition of misuse referred to the (1) use without a prescription of one's own; (2) use in greater amounts, more often, or longer than told to take; or (3) use in any other way doctors did not direct to use (Center for Behavioral Health Statistics and Quality, 2018).

2.2. Measures

The outcome variable of the current study was a binary measure of opioid misuse that includes misuse of prescription opioid pain relievers (such as hydrocodone, oxycodone, and morphine) and use of heroin (Substance Abuse and Mental Health Services Administration, 2018). The independent variables were extensively selected from all measured NSDUH variables that showed relations with adolescent substance use behavior in the literature (Becker et al., 2008; Edlund et al., 2015; Jones, 2018; Rigg and Monnat, 2015; Sullivan et al., 2006; Wu et al., 2008).

2.2.1. Sociodemographic characteristics

The following sociodemographic variables were included in the model: sex, race/ethnicity, class standing (grade 8 or lower, grades 9–12, and high school diploma), household income, the number of people living in the household, health insurance status including Medicaid, and overall health status.

2.2.2. Exposure to primary prevention programming and efforts

The following variables were also included in the model: participation (yes/no) in a special class about drugs or alcohol in school, alcohol/tobacco/drug prevention programs outside of school, programs or group counseling to deal with drug or alcohol use, pregnancy or sexually transmitted disease prevention programs, violence prevention programs, or problem solving/communication skills/self-esteem groups; having seen or heard any alcohol or drug prevention messages from sources outside school (yes/no); and having talked with parents about the dangers of tobacco/alcohol/drug use (yes/no).

2.2.3. Youth experience with school, parents, mental health service, and substance use

Youth experience with school was measured through the following variables: grade average for the last grading period completed (A, B or C vs. D or lower), overall feeling about going to school (liked a lot or kind of liked vs. didn't like very much or hated), and the number of school-based or non-school-based youth activity participation (none or one vs. two or more). Youth experience with parents was measured by the number of times one had a fight with a parent (9 or fewer vs. 10 or more), how often (always or sometimes vs. seldom or never) parents provide help with homework, and how often (always or sometimes vs. seldom or never) parents limit the amount of time going out with friends on school nights. The following items were used to measure mental health: having a major depressive episode in the past year (yes/no), having treatment/counseling from a therapist/psychologist/psychiatrist/social worker/counselor for emotional/behavioral problems that were not caused by alcohol or drugs (yes/no), and receiving non-

Table 1
Selected descriptive statistics of the sample aged 12–17 years: 2015–2017 NSDUH (N = 41,579).

Variables	Overall	Misuser (n = 1521)		Non-misuser (n = 40,058)		p-Value
	N	N	%	N	%	
Sociodemographics						
Sex						0.005
Male	21,246	692	3.3	20,554	96.7	
Female	20,333	829	3.9	19,504	96.1	
Race/ethnicity						< 0.001
Non-Hispanic White	22,182	771	3.4	21,411	96.6	
Non-Hispanic Black	5438	211	3.8	5227	96.2	
Non-Hispanic native American/Hawaiian ^a	852	32	3.0	820	97.0	
Non-Hispanic Asian	1601	38	2.2	1563	97.8	
Hispanic	9182	355	3.9	8827	96.1	
Others	2324	114	4.9	2210	95.1	
Class standing						< 0.001
Grade 8 or lower	22,489	507	2.2	21,982	97.8	
Grades 9–12	18,640	974	5.2	17,666	94.8	
High school diploma or above	450	40	9.4	410	90.6	
Overall health status						< 0.001
Excellent	14,126	359	2.5	13,767	93.6	
Very good	17,190	609	4.6	16,581	95.0	
Good	8519	437	5.0	8082	96.7	
Fair/poor	1732	116	6.4	1616	97.5	
Exposure to primary prevention programming and efforts						
Drug education in school						< 0.001
Participated	28,210	953	3.3	27,257	96.7	
Never participated	10,860	465	4.2	10,395	95.8	
Substance use program/counseling						< 0.001
Participated	1557	121	6.9	1436	93.1	
Never participated	39,344	1364	3.4	37,980	96.6	
Seen drug prevention message						0.041
Yes	29,669	1047	3.5	28,622	96.5	
No	11,095	438	3.9	10,657	96.1	
Talking to parents about danger of substance use						< 0.001
Yes	23,225	759	3.2	22,466	96.8	
No	17,525	726	4.1	16,799	95.9	
Youth experience						
Number of fights with parents						< 0.001
9 or fewer	32,853	972	2.9	31,881	97.1	
10 or more	7843	518	6.4	7325	93.6	
Major depressive episode experience						< 0.001
Yes	5354	437	8.0	4917	92.0	
No	35,064	1023	2.9	34,041	97.1	
Emotional treatment from therapist						< 0.001
Yes	4872	353	7.2	4519	96.9	
No	36,093	1130	3.1	34,964	92.8	
Tobacco use experience						< 0.001
Ever user	5342	781	14.4	4561	85.6	
Never user	36,237	740	2.9	35,497	97.9	
Alcohol use experience						< 0.001
Ever user	11,792	1129	9.4	10,663	90.6	
Never user	29,771	392	1.3	29,379	98.7	
Marijuana use experience						< 0.001
Ever user	6733	915	13.5	5818	86.5	
Never user	34,818	605	1.8	34,213	98.2	

Note: Frequencies are unweighted and percentages are weighted. Frequencies may not sum to the total due to missing observations. Differences between opioid misuser and non-misuser were tested via weighted Pearson's chi-squared tests. NSDUH = National Survey on Drug Use and Health; CHIP = children's state-assisted health insurance; STD = sexually transmitted disease; PREG = pregnancy.

^a Non-Hispanic native American/Hawaiian indicates non-Hispanic American Indian, Alaska native, native Hawaiian, or other Pacific islander.

specialty mental health services (e.g., foster care or juvenile detention center) (yes/no). Youth substance use experience was measured by ever smoking part or all of a cigarette (yes/no), ever having a drink of any type of alcoholic beverage (yes/no), and ever using marijuana or hashish (yes/no).

2.3. Model development

Prediction models for adolescent opioid misuse were developed via three different ML algorithms: artificial neural networks, distributed random forest, and gradient boosting machine. These algorithms were selected based on their advantages such as resiliency to overfitting,

modeling nonlinear associations, the ease of implementation, and general acceptance in the relevant scientific community of ML technique (Taylor et al., 2018). Each algorithm in a training dataset and application of the algorithms to a test dataset was developed through the open-source R programming language version 3.6.0 (R Core team, 2019) and the H2O (<http://www.h2o.ai>) R interface version 3.24.0.5. H2O offers a powerful Java-based platform that includes implementations of ML algorithms. A detailed description of the algorithms in H2O platform is found in H2O documentation (Aiello et al., 2017).

The dataset was randomly divided into a training set and a test set (70% vs. 30%) with respect to the outcome variable. After splitting the dataset, we fitted ML models in the training set using the five-fold cross-

validation. That is, the training dataset was partitioned into five subsets including four training sets and one validation set, and the model was fitted by estimating errors on the training sets and evaluated on the validation set. Through this repetitive procedure, the error estimations were averaged over all repetitions to obtain total effectiveness of a model. Then, we fitted ML models using the test dataset. ML algorithms typically have hyperparameters that define specific functions to be learned. For example, hyperparameters include the number of trees for tree-based algorithms (e.g., random forest), the number of layers in neural networks models, and the input/output data scaling. Optimal hyperparameter values were computed in a way described in the literature (Suchting et al., 2018). Full hyperparameter search space for each algorithm and final model hyperparameters are reported in Appendix A.

After examining descriptive statistics and distributions of study variables, we fitted ML prediction models of opioid misuse using three algorithms. Then, we fitted the same model using the logistic regression for comparison with ML approaches. Instead of maximum-likelihood estimation method, we used the penalized logistic regression to alleviate the bias resulting from the potential multicollinearity and the imbalance in the outcome variable. Missing data ($n = 7140$) were imputed using the multiple imputations with chained equations (MICE) technique in R with 5 imputed data sets created using 40 imputation iterations for each data set.

2.4. Statistical analysis

We used Pearson's chi-squared statistic to test the differences between opioid misuse and non-misuse. Model performance was evaluated in terms of the area under the receiver operating characteristic curve (AUROC) and the area under the precision-recall curve (AUPRC). The AUROC plots true positive rates against false positive rates for different cut-off points of a parameter (Bradley, 1997), displaying a trade-off between sensitivity and specificity over all possible threshold values. Unlike the AUROC, the AUPRC plots positive predictive values against true positive rates (Davis and Goadrich, 2006), displaying a trade-off between precision and sensitivity over all possible threshold values. Another difference between the two metrics is dependence on prevalence. Specifically, when prevalence is low, the expected value of random guessing of the AUPRC is close to 0 while that of AUROC is 0.5 (Boyd et al., 2013; Ozenne et al., 2015). We used the AUPRC as a primary measure of prediction performance given that it has been considered more informative for assessing binary classifiers on imbalanced outcome variable (Saito and Rehmsmeier, 2015). To examine relative importance of the explanatory variables in explaining opioid misuse among adolescents, we examined the relative influence of each variable in the best performing model.

3. Results

3.1. Descriptive statistics

Of the 41,579 adolescents aged 12 to 17 years in the study sample, 3.7% ($n = 1521$) misused opioids in the past year. As shown in Table 1, compared to adolescents who did not misuse opioids, those who misused showed significant differences in terms of sociodemographics, exposure to primary prevention programming and efforts, and youth experience. Further details of the sample characteristics are presented in Appendix B.

3.2. Results from machine learning and logistic regression

A prediction model was constructed for each of the three ML algorithms and for the penalized logistic regression analysis to compare model performance between the four different approaches. Table 2 shows performance results of the four approaches in predicting opioid

Table 2

Performance results of three different machine learning models and the penalized logistic regression in predicting opioid misuse among U.S. adolescents.

Model	AUROC	AUPRC
Artificial neural networks (threshold = 0.196)	0.810	0.157
Distributed random forest (threshold = 0.114)	0.809	0.172
Gradient boosting machine (threshold = 0.055)	0.811	0.160
Penalized logistic regression (threshold = 0.139)	0.815	0.162

Note: AUROC - area under the receiver operating characteristic curve, AUPRC - area under the precision-recall curve. Values in the parentheses represent the optimized probability thresholds that maximize the accuracy.

misuse among U.S. adolescents, including AUROC, AUPRC, and the optimized probability thresholds that maximize prediction accuracy for each model (see Appendix C for classification frequencies). In terms of AUROC, the penalized logistic regression was the best-performing model (0.815) followed by gradient boosting machine (0.811), artificial neural networks (0.810), and distributed random forest (0.809). In terms of AUPRC, the distributed random forest showed the best performance in prediction (0.172) followed by penalized logistic regression (0.162), gradient boosting machine (0.160), and artificial neural networks (0.157). Because AUPRC is the primary measure of prediction performance on this study with an imbalanced outcome variable (Saito and Rehmsmeier, 2015), the distributed random forest was deemed as the best-performing model. The true positive rate (i.e., proportion of correctly predicted opioid misusers) at the optimal threshold was approximately 40% on the distributed random forest. The plots of AUROC and AUPRC of the distributed random forest are presented in Figs. 1 and 2.

The relative influence of each variable in the distributed random forest (i.e., the best-performing model in terms of AUPRC) in explaining opioid misuse among adolescents is presented in Table 3. Marijuana experience was ranked to be the most important explanatory variable in explaining adolescent past-year opioid misuse followed by cigarette smoking and alcoholic beverage consumption. Those three substance use experiences were also ranked to be the top three important

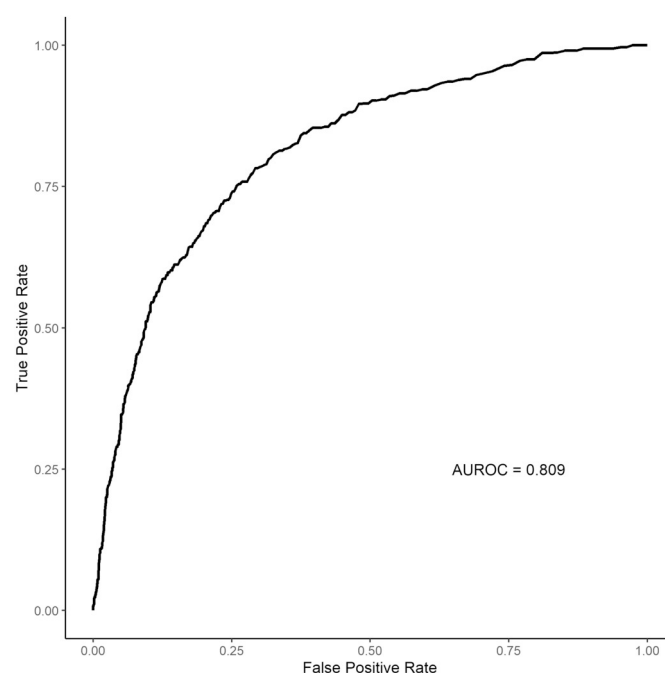


Fig. 1. Model performance of the best-performing model (distributed random forest) for opioid misuse among U.S. adolescents based on the area under the receiver operating characteristic curve (AUROC).

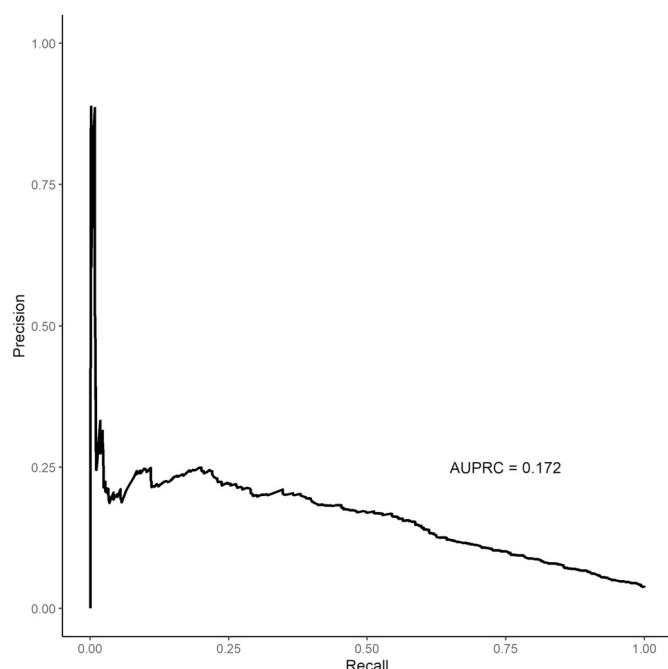


Fig. 2. Model performance of the best-performing model (distributed random forest) for opioid misuse among U.S. adolescents based on the area under the precision-recall curve (AUPRC).

Table 3

Importance of the variables in the best-performing model (distributed random forest) based on each variable's relative influence.

	Variable	Scaled importance
1	Marijuana use experience	1.000
2	Tobacco use experience	0.747
3	Alcohol use experience	0.425
4	Major depressive episode experience	0.137
5	Number of fights with parents	0.105
6	Race/ethnicity	0.068
7	Overall health status	0.066
8	Emotional treatment from therapist	0.063
9	Class standing	0.052
10	Feeling about going to school	0.052

Note: Importance of the variables was calculated based on the improvement in squared error (over all trees) given a node split on a variable during the tree building process.

variables in the gradient boosting machine and penalized logistic regression models, affirming face validity of the model (see Appendix D).

4. Discussion

This study contributes to the literature and the field as this is the first study that used ML techniques to develop a prediction model of opioid misuse in a nationally representative sample of U.S. adolescents. This investigation has been called for because ML offers a potential approach to build reliable and flexible predictive models based on computational learning and pattern recognition (Obermeyer and Emanuel, 2016). In the current study, even with a relatively small set of variables and cases, machine learning models showed good performance in predicting adolescent opioid misuse. Given that ML techniques often require a vast number of cases and variables to assure prediction/classification accuracy in tasks (Taniguchi et al., 2018; Walsh et al., 2017), the illustration of good prediction performance of a ML technique with a modest number of variables and typical sample size for a national surveillance dataset is novel and meaningful. This means that ML models do not necessarily have to be very complex and based

on a large sample size. Researchers could use ML methods for various national surveillance datasets that are publicly available.

The distributed random forest was the best performing model in predicting adolescent opioid misuse with the highest AUPRC, outperforming the penalized logistic regression. This suggests that ML techniques can be a promising technique in the prediction of adolescent opioid misuse. While AUROC is broadly used in the literature as an evaluation metric for binary classifiers, it can be misleading and less informative when data are highly imbalanced (Davis and Goadrich, 2006). For such an imbalanced outcome, AUPRC is deemed superior to AUROC in evaluating model performance (Chiew et al., 2019; Davis and Goadrich, 2006). Because AUPRC leads to more accurate interpretation of model performance than AUROC by displaying susceptibility of classifiers to highly skewed data (Chiew et al., 2019), future similar studies of imbalanced outcomes such as substance use behaviors are recommended to report both AUROC and AUPRC values. It should be also noted that literature indicates no clear consensus regarding the best choice of ML algorithm in terms of prediction performance (Thottakkara et al., 2016). Replication research in other settings, populations, and topical areas is warranted to evaluate and compare prediction performance of various ML techniques as well as conventional analytical methods to help build on evidence of relative prediction performance of different methods. Until such evidence is established, it seems appropriate to examine multiple ML techniques on the same model.

Given the rapid increase in drug overdose and misuse these days, it is increasingly important to implement an effective program to identify susceptible individuals and prevent them from misuse. Identification of predictors of adolescent opioid misuse in order of their relative influence in prediction in the best performing model offers additional insights into potential contributing factors to opioid misuse, which can help prioritize prevention efforts. The three variables with the largest predictive power were all related to substance use. Earlier studies already demonstrated that adolescents with risk behaviors tend to show an array of other risk behaviors (Schofield et al., 2008; Tapert et al., 2001). According to the Problem Behavior Theory, health risk behaviors tend to co-occur because they have common socioecological factors (Jessor, 1992). These empirical supports and theoretical explanations suggest that health professionals and researchers may need to tackle underlying factors associated with the propensity of engaging in multiple risk behaviors among adolescents, which may help reduce and prevent multiple problematic behaviors, including opioid misuse.

Typical conventional approaches to screening individuals who are at high risk for opioid misuse are costly and time-consuming especially in limited resource settings. Thus, ML approaches using a nationally representative data may be useful as they may provide a cost-effective screening strategy or a better target population segmentation in opioid misuse prevention efforts based on the strongest-predicting variables (Suchting et al., 2018; Hvistendahl, 2016). In addition, the optimized probability threshold that maximizes prediction accuracy of the best performing ML model can be utilized as a guide to screening or segmentation efforts by enabling risk-benefit assessment of potential screening/segmentation plans (Suchting et al., 2018). A caution is warranted, however, in interpreting this benefit of the prediction model. Given that it is a population-based rather than clinical-based approach, one should not infer opioid use on an individual level based on a set of predictors. Such an approach would be not only unethical but also misleading and erroneous (e.g., ecological fallacy). Rather, the information gleaned from this type of predictive model should be used at the population level and help health experts and professionals use limited prevention resources more effectively.

Another notable finding that deserves mention is that substance use experiences were ranked to be the top important variables irrespective of the algorithm used, affirming face validity of the prediction models of opioid misuse. Importantly, the penalized logistic regression model also ranked the same three variables as the most important variables

with interpretable coefficients. This is meaningful as it not only adds to face validity of the model but also complements one of the shortcomings of the tree-based ML models (e.g., random forest, gradient boosting) which are commonly considered as black-box type classifiers (Breiman, 2001), indicating a lack of interpretability. Adding interpretability to the ML prediction model can enhance the chances of adoption in practice immensely (Vellido, 2019). Therefore, it is recommended for researchers to include the penalized logistic regression model in the pool of ML models to be tested and compared.

It should be noted that there may be other important predictors for adolescent opioid misuse that were not measured by the NSDUH or were omitted from analysis even if they were measured. In addition, it is even plausible that some variables mask or partially represent the effect of other variables. For example, the literature indicates Medicaid enrollees are more likely than non-Medicaid counterparts to inappropriately use opioids (Centers for Medicare and Medicaid Services, 2016; Centers for Disease Control and Prevention, 2011). However, Medicaid is not within the top 10 variables of importance in prediction in the best performing model of the current study. Given that Medicaid enrollees are more likely to be tobacco and alcohol users than their counterparts (Naghavi et al., 2016; Weyh et al., 2015; El-Sayed et al., 2011), it is very likely that such substance use variables explain much of the variance of Medicaid. Thus, precautions need to be taken against misinterpretations of the list of the variables of importance in prediction. Also, more replication studies are needed to corroborate the findings of the current study.

This study had limitations. First, we used ML techniques provided by H2O platform. We cannot eliminate the possibility that other ML techniques might provide a better prediction performance. However, H2O platform not only offers highly optimized learning algorithms, but also provides model performance at different thresholds with a good summary. Thus, we believe that the ML techniques used for this study provided highly optimized ML algorithms to obtain superb, if not the best, prediction. Second, as in all nonexperimental studies, we cannot rule out the possibility of residual and unmeasured confounding. Third, as in all secondary analyses, the analyses were limited to the collected information in the NSDUH. It is plausible that some information that was not collected in the NSDUH may help enhance the prediction performance of the models examined in this study. Fourth, due to the cross-sectional design, causal interpretation cannot be made.

Despite these limitations, this study contributes to the literature as this is the first study that used ML techniques to develop a prediction model of opioid misuse in a nationally representative sample of U.S. adolescents and demonstrated that ML techniques can be a promising technique in the prediction of adolescent opioid misuse. Among the ML techniques examined, the distributed random forest showed the best performance. Future replication research in other settings, populations, and topical areas would be desirable that evaluates and compares prediction performance of various ML techniques as well as conventional analytical methods to help build on evidence of relative prediction performance of different methods.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ypmed.2019.105886>.

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